

GyuYeop Do

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Education

Seoul National University	March 2020 - July 2026
B.S. Electrical & Computer Engineering	Seoul, Republic of Korea
- Includes 1.5 years of mandatory military service as a Korean Augmentation to the United States Army (KATUSA) soldier - Relevant Coursework: Deep Learning, 3D Computer Vision, Computer Systems Programming, Linear Algebra	

Research Experience

Seoul National University - COSS Program	December 2025 - Present
SNU COSS Group (Advisor: Nam-Joon Kim)	Seoul, Republic of Korea
Conducting research on open-vocabulary segmentation for edge devices.	

Seoul National University - Electrical and Computer Engineering	September 2025 - December 2025
Machine Perception and Reasoning Lab (Advisor: Jonghyun Choi)	Seoul, Republic of Korea
Undergraduate Thesis: Enhancing action prediction in Vision-Language-Action (VLA) models with 3D foundation models.	

Seoul National University - Computer Science	December 2024 - July 2025
Visual & Geometric Intelligence Lab (Advisor: Jaesik Park)	Seoul, Republic of Korea
- Implemented key modules of an RGB-only grasp prediction pipeline based on normal-field estimation to improve robotic grasping of transparent objects. - Built an RGB-only grasp prediction pipeline leveraging feed-forward 3D feature inference and surface normal prediction for objects challenging for conventional 3D reconstruction methods. - Developed a method for reconstructing signed distance functions (SDFs) of dynamic rigid objects from images captured during motion.	

Experience

Team Gravity	March 2020 - Present
Founder	Seoul, Republic of Korea
- Founded and led an educational content business producing Korean SAT Physics study materials, publishing 6+ commercially sold prep books available in major bookstores, and selling content to one of South Korea's largest education enterprises. - Managed a team of ~20 physics content developers and 5 illustrators over 6 years, overseeing content development, quality control, and production for high-difficulty physics problem sets.	

Undergraduate Deep Learning Club

January 2025 - Present

Club President

Seoul, Republic of Korea

- Led an undergraduate deep learning club across three cohorts, each focused on an in-depth study track in Reinforcement Learning, Computer Vision, and NLP.
- Facilitated weekly paper readings, discussions, and project-based learning, with each cohort completing a final project.

AWS & Seoul National University - Data Venture Hackathon

July 2025 - Present

Team Lead

Seoul, Republic of Korea

- Team lead in the AWS Korea & SNU Data Venture Hackathon, developing a platform for franchise owners.
- Contributed to the majority of the codebase, implementing a multi-agent AI system, a backend built with Spring Boot, and a frontend developed using Next.js.
- Built a robust AI system with multiple agents and RAG to answer franchise owners' questions based on the brand's info and Korean legal documents.

POSTECH Big Data Challenge - Solar Irradiance Estimation Competition November 2025 - December 2025

Team Lead

Republic of Korea

- Conducted tabular data analysis to model solar irradiance using LightGBM and XGBoost with an ensemble approach.
- Conducted extensive feature engineering and data visualization to improve accuracy.

Qualcomm & Microsoft - Edge AI Hackathon

July 2025 - August 2025

Team Lead

Seoul, Republic of Korea

- Led development of a fully offline, real-time generative storytelling app on a Snapdragon-based Windows laptop, with a PySide desktop interface.
- Optimized Llama 3.2 3B for real-time on-device text generation by compiling it with the Qualcomm QNN SDK for NPU acceleration.
- Integrated CLIP and Stable Diffusion, compiling and running Stable Diffusion on the NPU via ONNX Runtime for efficient edge inference.

AWS & Seoul National University - Solving a Social Problem Hackathon

January 2025 - Present

Main Developer

Seoul, Republic of Korea

- Developed an Android application using Jetpack Compose integrated with AWS services.
- Implemented an AI-driven system using RAG and speech-to-text (STT) to automatically warn users of potential scams or fraudulent content immediately after messages or phone calls.

Republic of Korea Army - Korean Augmentation to the United States Army

November 2022 - May 2024

Pyeongtaek, Republic of Korea

- Completed 1.5 years of mandatory military service as Korean Augmentation to the United States Army (KATUSA), serving in the 1st Signal Brigade, U.S. Army.

Teaching

Sidae Injae Academy - Korean SAT Physics Prep

January 2019 - July 2019

TA for Instructor Song-Do Kim

Seoul, Republic of Korea

- Supported Song-Do Kim, a recognized instructor with 1,000+ students in large-scale offline lectures.
- Developed core instructional materials, including problem sets, quizzes, assignments used across classes.

Sidae Injae Academy - Advanced High School Mathematics

January 2019 - July 2019

TA for Instructor Young-Seok Park

Seoul, Republic of Korea

- Assisted Young-Seok Park in teaching classes of 100+ students by providing one-on-one explanations for advanced problem-solving and creating textbooks and assessment materials.

Awards

Army Commendation Medal (U.S. Army)

July 2024

Awarded by HQ Eighth Army during service in the 1st Signal Brigade, U.S. Army

Approved by Brigadier General Roderick F. Laughman (USA)

Recognized for exceptional service and contribution during military duty in coordination with the U.S. Army.

Exemplary KATUSA Commendation

February 2024

Awarded by 2nd Area Commander, Republic of Korea Army (ROKA) Support Group Lieutenant Colonel Kim, Ki-Hyun (ROKA)

Recognized for exceptional professionalism, dedication, and exemplary military conduct. Demonstrated leadership, discipline, and performance that set a high standard within the KATUSA (Korean Augmentation to the United States Army) program.

AWS & Seoul National University - Data Venture Hackathon

October 2025

Top 6 Finalist - advanced to the final round (award pending).

Pohang University of Science and Technology - Big Data Challenge

December 2025

Bronze award. Ranked top 7 of 140 participants.

Qualcomm & Microsoft - Edge AI Hackathon

August 2025

4th Place Award.